# ⚙️ Asset Failure Prediction using Machine Learning

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## 1. Introduction

In modern industries, machines are essential for continuous production. Unexpected machine failures can lead to downtime, increased costs, and reduced efficiency. To address this issue, predictive maintenance techniques are used to anticipate failures before they occur.

This project focuses on building a machine learning model that predicts whether a machine is likely to fail based on operational parameters such as temperature, rotational speed, torque, and tool wear.

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## 2. Problem Statement

The objective of this project is to develop a system that can predict machine failure in advance using historical data. This helps in performing maintenance proactively rather than reactively, thereby reducing downtime and improving system reliability.

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## 3. Dataset Description

The dataset used in this project contains machine-related parameters, including:

- Air temperature
- Process temperature
- Rotational speed
- Torque
- Tool wear
- Machine failure (target variable)

The dataset represents real-world inspired industrial sensor data and is used to train the model to identify patterns leading to machine failure.

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## 4. Methodology

### 4.1 Data Preprocessing

- Removed unnecessary columns such as UDI and Product ID
- Converted categorical variables into numerical format using encoding
- Separated features and target variable

### 4.2 Model Selection

A **Random Forest Classifier** was used for this project due to:

- High accuracy
- Ability to handle complex patterns
- Robustness against overfitting

### 4.3 Training and Testing

- The dataset was split into training and testing sets (80:20 ratio)
- The model was trained on the training data
- Predictions were made on the test data

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## 5. Results

The model was evaluated using accuracy score. The results showed that the model was able to predict machine failure with good accuracy.

Additionally, feature importance analysis helped identify which factors contributed most to machine failure, such as tool wear and torque.

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## 6. Challenges Faced

- Difficulty in obtaining a real-world dataset
- Handling categorical data
- Understanding feature importance
- Working with a small dataset

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## 7. Applications

This project can be applied in:

- Manufacturing industries
- Automotive systems
- Aircraft maintenance
- Power plants

It helps in reducing downtime, saving costs, and improving operational efficiency.

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## 8. Conclusion

This project successfully demonstrates how machine learning can be used for predictive maintenance. By analyzing machine parameters, the model can predict failures in advance, allowing timely intervention.

Although the dataset used is simplified, the project provides a strong foundation for real-world applications and can be extended using real-time data.

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## 9. Future Scope

- Integration with real-time IoT sensor data
- Development of a user interface
- Use of larger and more complex datasets
- Implementation of advanced models like deep learning

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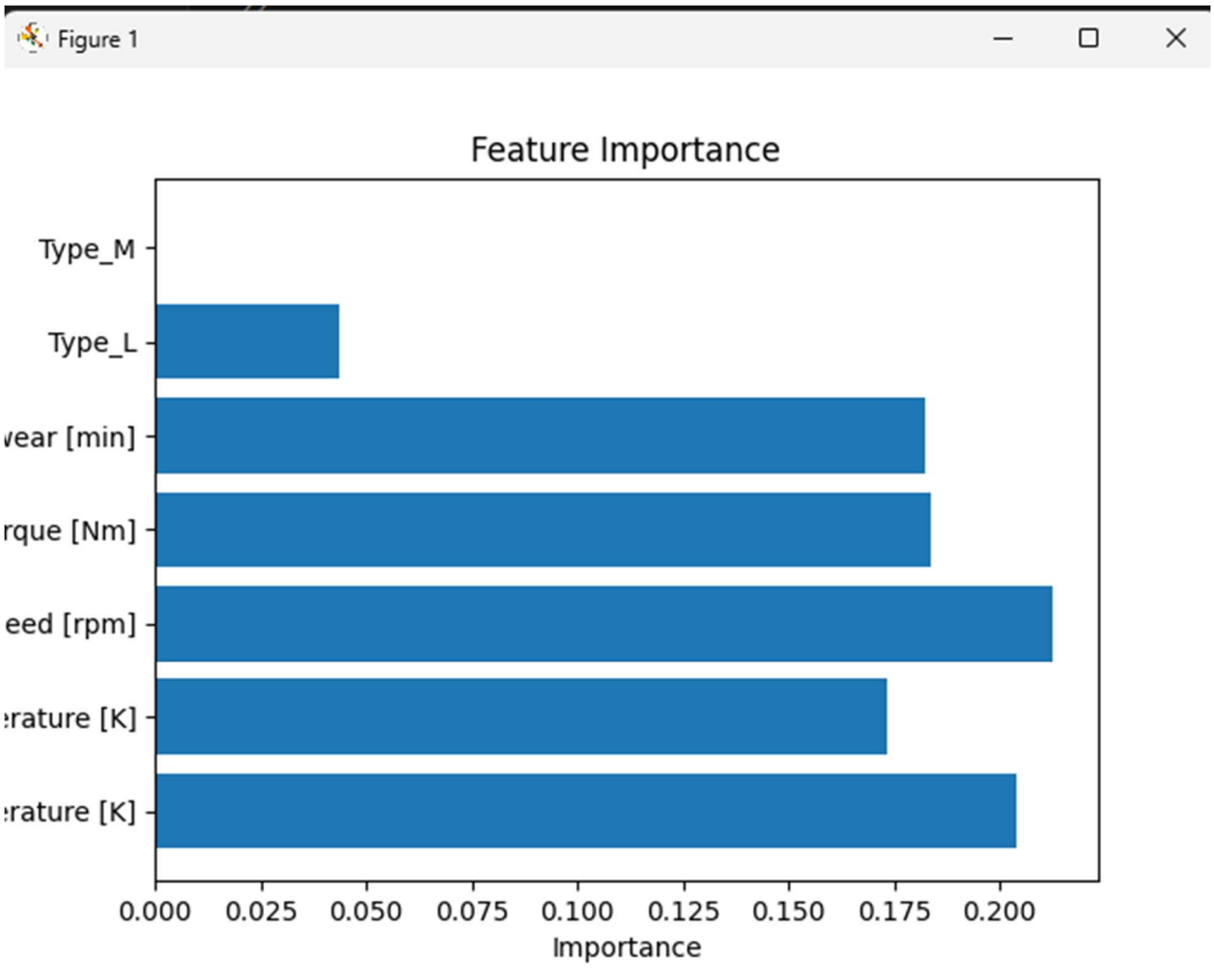
## 10. Learning Outcomes

- Understanding of machine learning workflow
- Data preprocessing techniques
- Model training and evaluation
- Practical application of predictive maintenance concepts

👩‍💻 **Author:** Pratibha Jha

# Output:




```
PROBLEMS 10   OUTPUT   ...   Filter (e.g. text, !excludeTe...   Code
Predictor\Asset_Failure_Predictor.py"
Accuracy: 0.5
Prediction (1=Failure, 0=Safe): 1
Failure Probability: 0.98

[Done] exited with code=0 in 82.284 seconds
```